MULTIVARIABLE CALCULUS • UNIT 2.01

C Multivariable Chain Rule, Directional Derivatives & Gradients

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 C	Q3 B	Q4 A	Q5 D
Q6 C	Q7 B	Q8 C	Q9 A	Q10 B
Q11 C	Q12 D	Q13 B	Q14 D	Q15 A
Q16 D	Q17 A	Q18 C	Q19 A	Q20 B
Q21 B	Q22 A	Q23 D	Q24 A	Q25 C

Multivariable Chain Rule, Directional Derivatives & Gradients Practice Exercises

1 MCQ

Find $\nabla f(-5, 3)$ for $f(x, y) = 4x^2 + 3y^2$.

A (18, -40)

B (-20, 9)

C **(-40, 18)** ✓

D (40, -18)

WORKED SOLUTION & VERIFICATION

$\nabla f = (8x, 6y)$, so $\nabla f(-5, 3) = (-40, 18)$.

2 MCQ

Find $\nabla f(-6, 3)$ for $f(x, y) = 3x^2 + y^2$.

A (6, -36)

B (-18, 3)

C **(-36, 6)** ✓

D (36, -6)

WORKED SOLUTION & VERIFICATION

$\nabla f = (6x, 2y)$, so $\nabla f(-6, 3) = (-36, 6)$.

3 MCQ

Find $\nabla f(5, -1)$ for $f(x, y) = 3x^2 + 5y^2$.

A (-10, 30)

B **(30, -10)** ✓

C (15, -5)

D (-30, 10)

WORKED SOLUTION & VERIFICATION

$\nabla f = (6x, 10y)$, so $\nabla f(5, -1) = (30, -10)$.

4 MCQ

Find $\nabla f(-5,-5)$ for $f(x,y) = 9x^2 + 6y^2$.

- **A (-90, -60)** ✓
- B (-60, -90)
- C (-45, -30)
- D (90, 60)

WORKED SOLUTION & VERIFICATION

$\nabla f = (18x, 12y)$, so $\nabla f(-5,-5) = (-90,-60)$.

5 MCQ

Find $\nabla f(-2,2)$ for $f(x,y) = 2x^2 + 7y^2$.

- A (28, -8)
- B (-4, 14)
- C (8, -28)
- **D (-8, 28)** ✓

WORKED SOLUTION & VERIFICATION

$\nabla f = (4x, 14y)$, so $\nabla f(-2,2) = (-8,28)$.

6 MCQ

Find $\nabla f(-2,-1)$ for $f(x,y) = 4x^2 + 8y^2$.

- A (-8, -8)
- B (16, 16)
- **C (-16, -16)** ✓
- D (-16, -14)

WORKED SOLUTION & VERIFICATION

$\nabla f = (8x, 16y)$, so $\nabla f(-2,-1) = (-16,-16)$.

7 MCQ

Find $\nabla f(-2,-4)$ for $f(x,y) = 2x^2 + 6y^2$.

- A (-48, -8)
- **B (-8, -48)** ✓
- C (-4, -24)
- D (8, 48)

WORKED SOLUTION & VERIFICATION

$\nabla f = (4x, 12y)$, so $\nabla f(-2,-4) = (-8,-48)$.

8 MCQ

Find $\nabla f(-1,-4)$ for $f(x,y) = 4x^2 + 8y^2$.

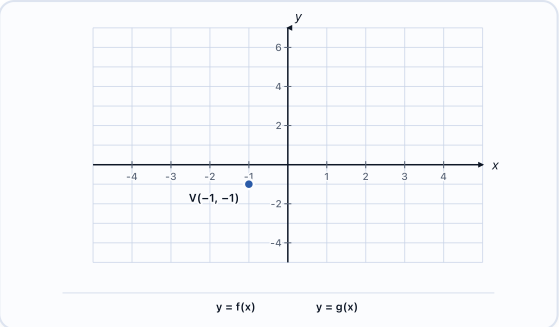
- A (-64, -8)
- B (-4, -32)
- **C (-8, -64)** ✓
- D (8, 64)

WORKED SOLUTION & VERIFICATION

$\nabla f = (8x, 16y)$, so $\nabla f(-1,-4) = (-8,-64)$.

9 MCQ

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.



- **A** 26 ✓
- **B** 10
- **C** 13
- **D** 52

WORKED SOLUTION & VERIFICATION

We are given $z = f(x, y) = x^2 + y^2$, with $x = 2t$ and $y = 3t$. We need to find $\frac{dz}{dt}$ using the Multivariable Chain Rule:
$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$.
First, find the partial derivatives of z with respect to x and y:
$\frac{\partial z}{\partial x} = 2x$
$\frac{\partial z}{\partial y} = 2y$
Next, find the derivatives of x and y with respect to t:
$\frac{dx}{dt} = 2$
$\frac{dy}{dt} = 3$
Now, substitute these into the chain rule formula:
$\frac{dz}{dt} = (2x)(2) + (2y)(3) = 4x + 6y$.
We are asked to find $\frac{dz}{dt}$ when $t = 1$. First, find the values of x and y when $t = 1$:
$x = 2(1) = 2$
$y = 3(1) = 3$
Finally, substitute these values of x and y into the expression for $\frac{dz}{dt}$:
$\left.\frac{dz}{dt}\right|_{t=1} = 4(2) + 6(3) = 8 + 18 = 26.$

10 MCQ

Consider the vector field $\mathbf{f}(x, y, z) = \langle 12x,\ 20y,\ 11z \rangle$ on R^3. Determine the scalar divergence $\nabla \cdot \mathbf{f}$.

- **A** 23
- **B** 43 ✓
- **C** 47
- **D** 55

WORKED SOLUTION & VERIFICATION

The divergence is $12 + 20 + 11 = 43$.

11 MCQ

For $f(x, y) = 8x - 15y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 23

B 16

C 17 ✓

D 18

WORKED SOLUTION & VERIFICATION

$\nabla f = (8, -15)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(8)^2 + (-15)^2} = 17$.

12 MCQ

For $f(x, y) = 4x - 3y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 7

B 4

C 6

D 5 ✓

WORKED SOLUTION & VERIFICATION

$\nabla f = (4, -3)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(4)^2 + (-3)^2} = 5.$

13 MCQ

For $f(x, y) = 8x - 6y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 14

B 10 ✓

C 9

D 11

WORKED SOLUTION & VERIFICATION

$\nabla f = (8, -6)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(8)^2 + (-6)^2} = 10.$

14 MCQ

For $f(x, y) = 24x - 7y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 31

B 24

C 26

D 25 ✓

WORKED SOLUTION & VERIFICATION

$\nabla f = (24, -7)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(24)^2 + (-7)^2} = 25$.

15 MCQ

For $f(x, y) = -20x + 21y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

- **A** 29 ✓
- **B** 41
- **C** 28
- **D** 30

WORKED SOLUTION & VERIFICATION

$\nabla f = (-20, 21)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(-20)^2 + (21)^2} = 29$.

16 MCQ

For $f(x, y) = -15x + 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

- **A** 23
- **B** 16
- **C** 18
- **D** 17 ✓

WORKED SOLUTION & VERIFICATION

$\nabla f = (-15, 8)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(-15)^2 + (8)^2} = 17$.

17 MCQ

For $f(x, y) = -6x - 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

- **A** 10 ✓
- **B** 14
- **C** 9
- **D** 11

WORKED SOLUTION & VERIFICATION

$\nabla f = (-6, -8)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(-6)^2 + (-8)^2} = 10$.

18 MCQ

For $f(x, y) = -3x - 4y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

- **A** 6
- **B** 7
- **C** 5 ✓
- **D** 4

WORKED SOLUTION & VERIFICATION

$\nabla f = (-3, -4)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(-3)^2 + (-4)^2} = 5$.

19 MCQ

For $f(x, y) = 12x - 5y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

- **A** 13 ✓
- **B** 12
- **C** 14
- **D** 17

WORKED SOLUTION & VERIFICATION

$\nabla f = (12, -5)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(12)^2 + (-5)^2} = 13$.

20 MCQ

For $f(x, y) = -9x + 40y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

- **A** 40
- **B** 41 ✓
- **C** 42
- **D** 49

WORKED SOLUTION & VERIFICATION

$\nabla f = (-9, 40)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(-9)^2 + (40)^2} = 41$.

21 MCQ

For $f(x, y) = -4x + 3y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

- **A** 7
- **B** 5 ✓
- **C** 4
- **D** 6

WORKED SOLUTION & VERIFICATION

$\nabla f = (-4, 3)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(-4)^2 + (3)^2} = 5.$

22 MCQ

For $f(x, y) = -3x + 4y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

- **A** 5 ✓
- **B** 7
- **C** 4
- **D** 6

WORKED SOLUTION & VERIFICATION

$\nabla f = (-3, 4)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(-3)^2 + (4)^2} = 5.$

23 MCQ

For $f(x, y) = 12x + 35y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 47 | B 36 | C 38 | D 37 ✓

WORKED SOLUTION & VERIFICATION

$\nabla f = (12, 35)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(12)^2 + (35)^2} = 37$.

24 MCQ

For $f(x, y) = -9x - 40y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 41 ✓ | B 40 | C 42 | D 49

WORKED SOLUTION & VERIFICATION

$\nabla f = (-9, -40)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(-9)^2 + (-40)^2} = 41$.

25 MCQ

For $f(x, y) = -12x - 35y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 47 | B 36 | C 37 ✓ | D 38

WORKED SOLUTION & VERIFICATION

$\nabla f = (-12, -35)$ everywhere. By Theorem 2.1, the maximum directional derivative equals $\|\nabla f\| = \sqrt{(-12)^2 + (-35)^2} = 37$.